

VentureIQ — System Architecture Specification

AI-Powered Multi-Agent Startup Due Diligence & Validation Engine

1. Problem Statement & Solution

Early-stage founders face major hurdles: generic LLMs give uncritical answers, manual due diligence is slow, and informal checks lack repeatable rigor. VentureIQ solves this through interactive discovery, Pinecone vector RAG retrieval, parallel LangGraph multi-agent analysis, and an Investor Readiness Index.

2. Tech Stack Matrix

Component	Technology	Role in VentureIQ
Frontend	React 19, Vite 7, Tailwind CSS 4	Responsive SPA workspace & dashboard
3D Core	Three.js r185, @react-three/fiber	Animated canvas reacting to state machine
Backend API	FastAPI, Uvicorn, Python 3.10	Async server with deduplication locks
Orchestration	LangGraph, LangChain	Stateful multi-agent workflow graph
LLM Model	Google Gemini (2.5 / 3.6 Flash)	Discovery chat & agent analytical reasoning
Vector DB (RAG)	Pinecone (rag-main), text-embedding-004	768-dim semantic evidence retrieval
Web Research	Tavily Search API	Real-time live competitor market search

3. LangGraph Workflow Execution

- **Phase 1: Discovery & Clarification Loop** — Lead Supervisor interviews founder, collecting structured context facts.
- **Phase 2: RAG Retrieval Node** — Generates 768-dim vector embedding and fetches top-k (k=5) documents from Pinecone index 'rag-main'.
- **Phase 3: Parallel Agent Execution** — Market (TAM), Competitor (Moat + Tavily search), Business (Economics), and Risk (Regulatory) agents run.
- **Phase 4: Synthesis & Scoring** — Executive Report agent synthesizes overall verdict and calculates 0-100 Investor Readiness scores.